

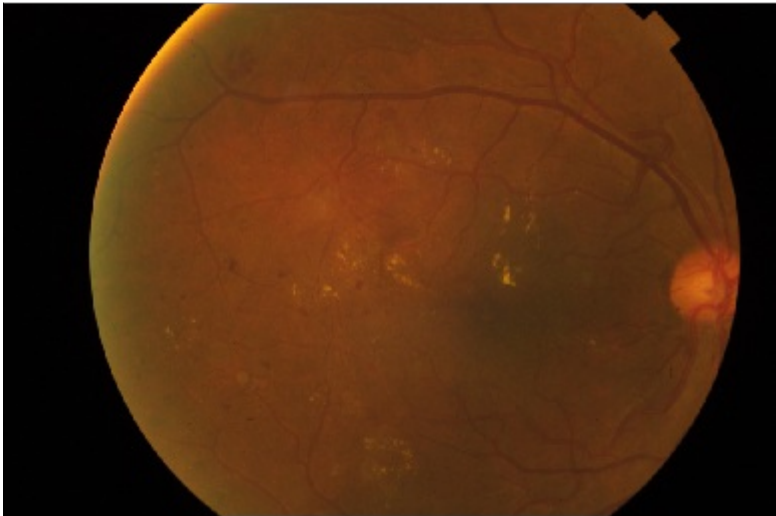
SIH 26038 — Explainable AI Diabetic Retinopathy Screening Report

Image ID: sample_fundus
Screening Date: 09-Sep-2026 07:19:51
Image Quality: BORDERLINE (Score: 2/3)
Sharpness: 19.3 | Brightness: 51.3 | Contrast: 31.3

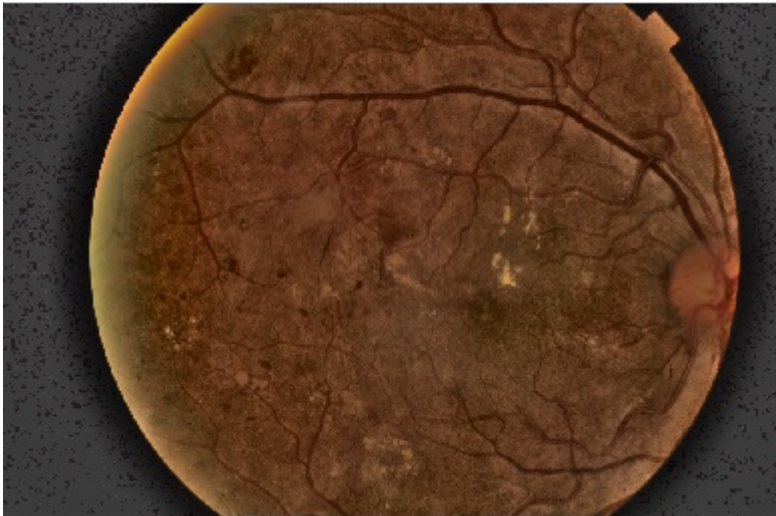
AI DIAGNOSIS: Moderate NPDR
Confidence: 92.8%
DR Severity Level: 2 / 4

URGENT REFERRAL — Tele-consult ophthalmologist within 7 days

Original Fundus Image



Enhanced (Quality: BORDERLINE)

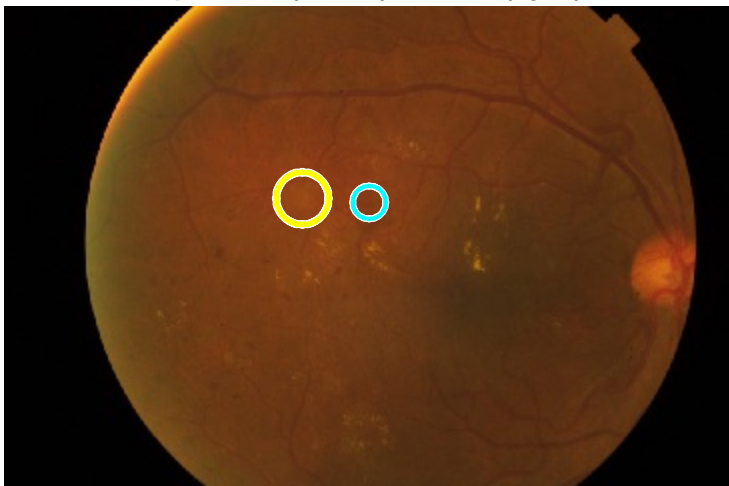


CLINICAL LESION EVIDENCE:
Microaneurysms (MA): 24 lesions (Earliest DR sign)
Hard Exudates (EX): 1 deposits (Lipid leakage)
Hemorrhages (HE): 0 blotches (Vascular damage)

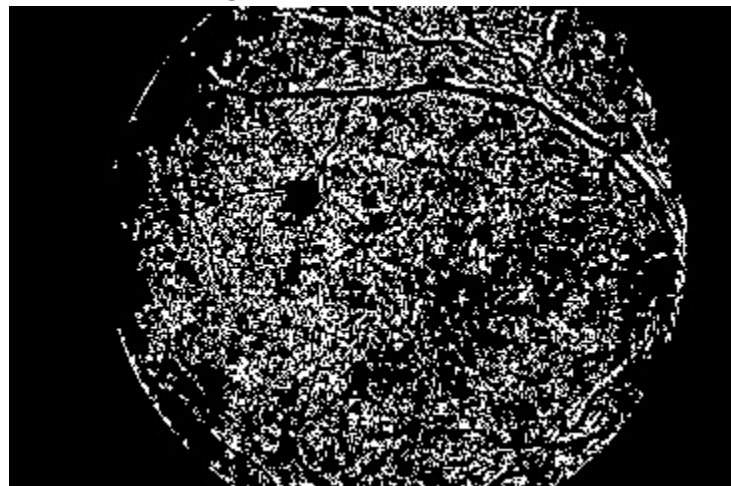
Optic Disc Center: [371, 239]
Fovea Center: [454, 244]

Retinal Structure Segmentation & Lesion Detection

Optic Disc (Yellow) & Fovea (Cyan)



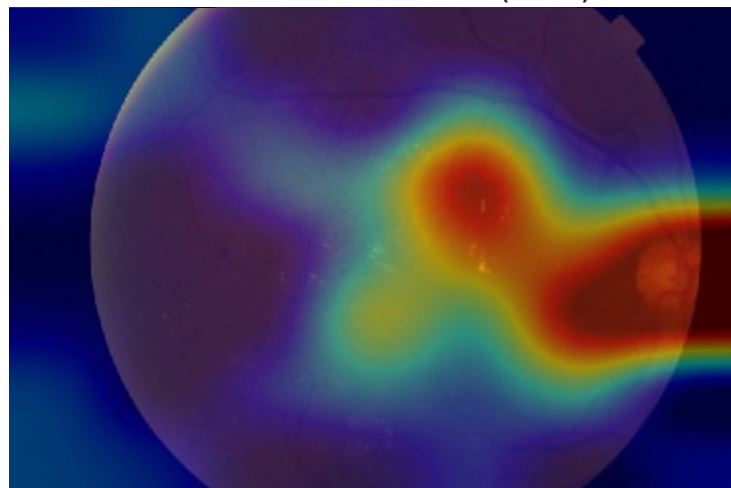
Segmented Vascular Network



MA(24) EX(1) HE(0)

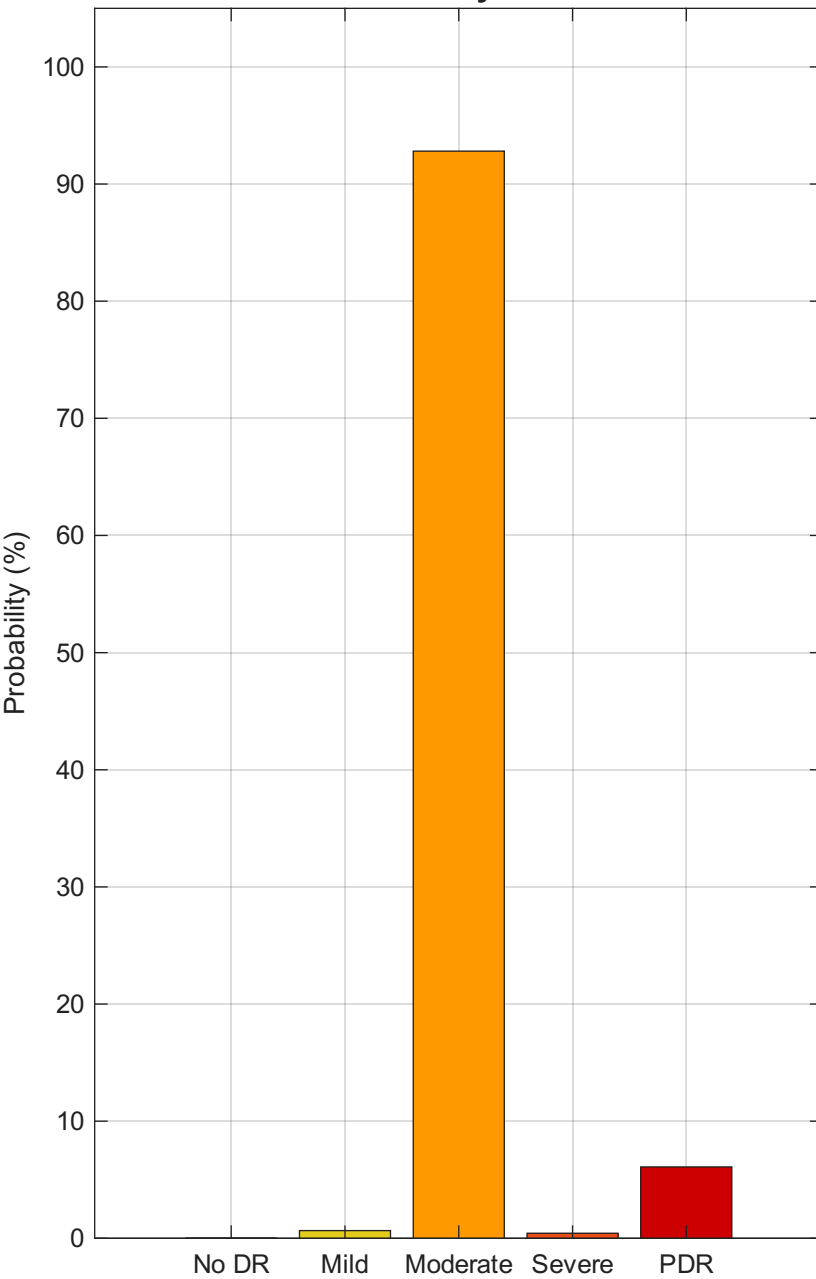


Grad-CAM: Moderate NPDR (92.8%)



AI Confidence Analysis & Class Probability Distribution

Class Probability Distribution



Clinical Decision Summary

CLINICAL TRIAGE DECISION

Predicted Class: Moderate NPDR

Confidence: 92.8%

Referable DR (Level 2+): true

Probability Breakdown:

No DR: 0.00%

Mild NPDR: 0.66%

Moderate NPDR: 92.81%

Severe NPDR: 0.43%

Proliferative: 6.10%

Validation: Human-in-the-loop review <30s